Magnetization of the Perry Formation of New Brunswick, and the rotation of Newfoundland

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The remanent magnetization of the Upper Devonian Perry Formation of New Brunswick consists of two components *B* and *A* acquired before and after folding (pre-Westphalian D) respectively. The stability spectra of these components overlap and it is therefore difficult to define them accurately and calculate their errors. The best estimate of the older (*B*) component (with respect to bedding) is 175, +23 (pole 32° N, 118° E), but this may be in error by 15° or more: its age is considered to be Late Devonian or Mississippian. The best estimate of the younger (*A*) component (with respect to present horizontal) is also 175, +23 and its age is considered to be early Pennsylvanian or older. The uncertainty in estimating these directions, however, means that the paleomagnetic observations, although they do not preclude the possibility that Newfoundland has rotated, also do not provide independent evidence that this has occurred, as has been suggested.

Introduction

The magnetization of the predominantly sedimentary Perry Formation of New Brunswick (previously studied by Black, 1964) has been studied from two localities (Fig. 1): St. Andrews, where the beds have low dips, and Blacks Harbour, where they dip more steeply. Its age of deposition has been determined as Upper Devonian (Alcock 1960) on the basis of plant fossils found at Blacks Harbour. The folding is considered to be pre-Westphalian D (D. G. Kelly, personal communication). The difference of tilt at the two localities allows a fold test to be made and at St. Andrews intraformational conglomerates (sites C1, C2, and C3) and interbedded lavas [two basalt flows (L1 and L2) and one rhyolite flow (L3)], with a baked contact (K), allow other field tests to be carried out. Therefore, at first sight the geological conditions appear ideal for obtaining an accurate paleomagnetic result with good control of the stability of magnetization and its age. Unfortunately the results turned out to be very complex, the magnetization dating from at least two periods in the history of the rock. The results nevertheless are of interest because they give another example of the difficulties encountered in obtaining paleomagnetic results from even mildly deformed rocks in orogenic zones and because they cast doubt on the paleomagnetic case for the rotation of Newfoundland.

Igneous Results

In the present work pilot specimens from each of 15 sites were thermally demagnetized by heating to successively higher temperatures and cooling in zero field (Robertson 1963). The intensity data are given in Table I. Bulk thermal cleaning of igneous rocks was carried out by heating to 500 °C (rhyolite to 400 °C). The results from the lava flows, the conglomerates, and the baked contact are now considered in turn, and the mean site directions are set out in Table II. The particularly interesting results of thermal demagnetization of the red sediments are described in the following section.

Lava Flows

The Perry basalts (sites L1 and L2) are magnetically unstable, their intensities falling rapidly on demagnetization (Table II). Furthermore there is very poor agreement (50° difference of mean direction after heating to 500 °C) between the two sites. Directions from the rhyolite (L3) are random ($P = 0.05$, Vincenz and Bruckshaw 1960) after heating to 400 °C.

Conglomerates

Rhyolite cobbles derived from lava flows in the lower part of the Perry sequence and obtained from conglomerates at sites C1 and C3, give widely scattered initial directions (Table II). After heating to 500 °C the directions are still scattered, but not random, and have a southerly tendency like the direction in the sediments (Fig. 2). The changes between the NRM and cleaned directions are consistent with the removal of a viscous component (Thellier 1937) directed along the present field. The basalt cobbles (C2) gave widely scattered directions, but their intensities are two orders of magnitude less than the

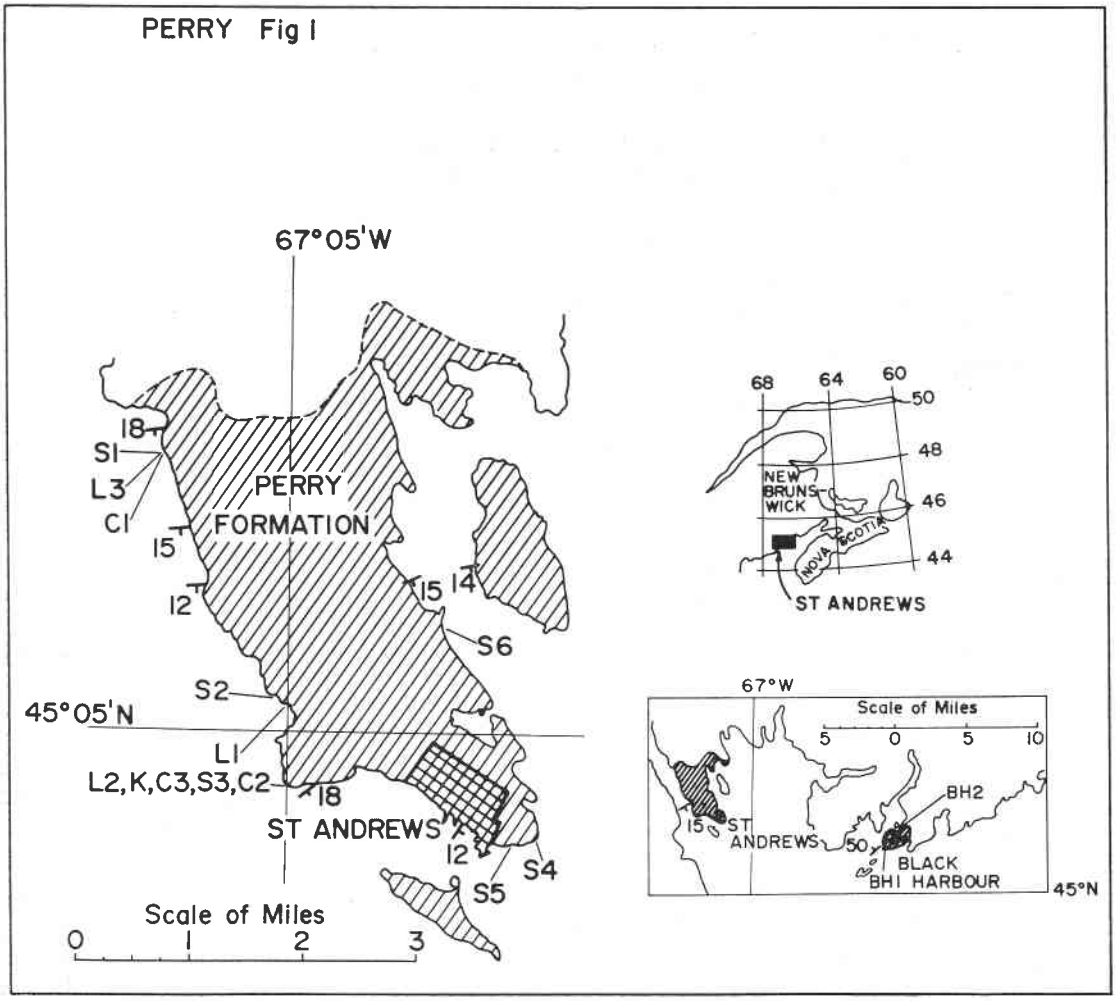


FIG. 1. Locality map showing the sampling sites on the St. Andrews peninsula and at Blacks Harbour, New Brunswick.

intensities found in the basalts *in situ* (Table I; sites L1 and L2). Thus the results from the rhyolite boulders show that the greater part of their magnetization was acquired after the conglomerates were formed, and the result from the basalt shows that either they were not derived from the Perry lavas or they have lost most of their original magnetization. In neither instance is there any evidence that the parent material is stable. This agrees with the data from the lavas themselves, therefore no reliable paleomagnetic information has been obtained from these lavas.

Baked Contact

At site K1 sandstones are baked to a gray color at the lower contact of a basalt lava. It is reasonable to suppose that the contact would acquire a magnetization (either TRM or CRM) during cooling (i.e. before folding). Although the intensity ratio is only 0.17 (Table I), the difference between the initial and cleaned (500 °C) directions is small (11°). This indicates that a small and thermally stable magnetization is present. The intensities remaining after cleaning at higher temperatures were too small to be reliable. The direction at 500 °C (160°, +21°) differs by 17° from the mean direction of the St. Andrews sediments (175°, +11°) discussed below. This is not regarded as significant considering that the baked sediment direction is

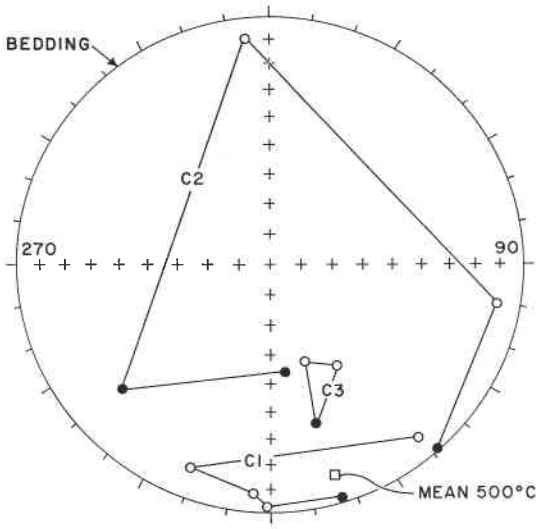


FIG. 2. Directions of magnetization in lava cobbles from conglomerates after heating samples to 500 °C. Combined mean for the rhyolite cobbles (C1 and C3) is also given. C2 is from basalt cobbles. Equal-area projection. Dots (circles) indicate positive (negative) inclinations.

only a spot reading of the field and does not average out the secular variation.

Red Sandstones

The sediments are current-bedded shallow water red sandstones and siltstones that were deposited quickly but intermittently. Therefore, although one site may represent only a short time span, many thousands of years may be represented by all the sites, so that the secular variation may be averaged out. If the magnetization is a chemical process (CRM) formed during diagenesis (Collinson 1965; Roy and Robertson 1968), the time taken for the magnetization to be acquired is not known, but it is likely to be thousands of years, and this would further enhance the coverage. The dip of the sandstones at St. Andrews is gently to the south-southeast, ranging from 10° to 18°. The strata at sites BH1 and BH2, in the Blacks Harbour area, dip to the northwest at about 50°. A fold test on the mean directions of these two groups, after demagnetization at 500 °C, is shown in Fig. 3. The differences (Table III) when the directions are referred to the present horizontal increase with thermal demagnetization, whereas they decrease when referred to bedding. Therefore the magnetization was acquired partly before and partly either during or after the folding, which is pre-Westphalian D. The differences, which are largely in inclination (Table II), are displayed in the vertical plane bearing 355° in Fig. 4. To facilitate comparison the two diagrams on Fig. 4 have been drawn with the bedding planes horizontal across the page. The bedding strikes are at an angle to the plane of the diagram and a cosine correction has been made to give the apparent dip in this plane. Because declinations are sufficiently close, no correction for magnetic dip was necessary. The resultants are plotted using the mean ratios listed in Table I and the mean inclinations given in Table II.

Demagnetization of a single component would give a reduction in intensity only, without direction change, but systematic direction changes do occur at Blacks Harbour (Fig. 4*b*) and it is therefore necessary to suppose that more than one component is present. In Fig. 4 we have assumed that two components (*B* and *A*) were acquired parallel to the geomagnetic field before

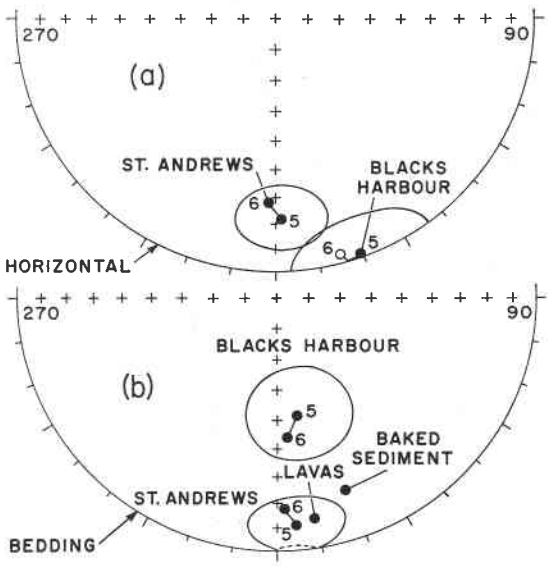


FIG. 3. (*a*) Numerals denote temperature of thermal cleaning in 100's of °C. Mean directions relative to the present horizontal, giving samples unit weight, for the sediments from St. Andrews and Blacks Harbour after thermal cleaning at 500 and 600 °C. The ovals of confidence at P = 0.05 are given for the means after the 500 °C treatment. Equal-area projection. Dots (circles) indicate positive (negative) inclinations. (*b*) Mean directions relative to bedding giving samples unit weight. The ovals of confidence are given for the sediments after the 500 °C treatment. Dots indicate positive inclinations.

TABLE I

Thermal demagnetization of pilot specimens

Site	No. of specimens	Initial intensity (e.m.u. $\times 10^{-6}$)	Ratio of intensity at this temperature to initial intensity						
			200 °C	300 °C	400 °C	450 °C	500 °C	550 °C	600 °C
S1	2	6.5	1.13	1.10	0.93	0.82	0.62	0.42	0.22
S2	1	13.7	1.02	0.94	0.36	0.21	0.13	0.14	0.11
S3	1	11.9	0.21	0.19	0.13	0.30	0.11	0.06	0.07
S4	2	5.0	0.94	0.86	0.70	0.64	0.50	0.26	0.19
S5	2	6.0	0.85		0.63		0.60		0.46
S6	2	5.6	0.69		0.44		0.31		0.27
Mean S1–S6							0.38		0.22
BH1	3	5.6	1.02		0.79		0.57		
BH1	1	6.4	0.81	0.80	0.66	0.61	0.47	0.34	0.36
BH2	3	5.0	1.00		0.68		0.64		
BH2	1	5.5	0.84	0.98	0.75	0.75	0.60	0.55	0.46
Mean BH1+2							0.59		0.41
K	3	35.2	1.11		0.29		0.17		
K	1	42.0	1.13	0.78	0.22	0.19	0.10		
C1*	1	8.6	0.94	0.41	0.57	0.42	0.45	0.21	0.21
C3*	3	27.2	1.00		0.91		0.79		
C2†	4	2.9			0.49		0.54		
L1†	4	122.1			0.25		0.07		
L2†	5	616.0			0.11		0.06		
L2†	1	850.0	0.16	0.06	0.03	0.03	0.02	0.01	0.01
L3*	1	2.3	0.74	1.04	1.17	0.61	0.61	0.78	0.74

*Samples of rhyolite
†Samples of basalt

TABLE II

Mean site statistics

Site	Initial					Heated to 500 °C					Heated to 600 °C				
	D_h I_h	D_b I_b	N	R	α	D_h I_h	D_b I_b	N	R	α	D_h I_h	D_b I_b	N	R	α
Red sandstones															
St. Andrews															
S 1	177, +44	175, +27	8	6.214	32	184, +32	181, +16	6	3.966	55	184, +17	183, +04	6	5.476	23
S 2	163, +19	161, +10	5	4.746	20	177, +24	173, +17	4	3.691	31	189, +26	185, +21	3	2.697	52
S 3	165, +38	160, +24	5	4.920	11	174, +37	161, +28	4	3.834	22	211, +59	187, +52	5	3.085	72
S 4	180, +34	177, +22	10	9.621	10	190, +12	189, +03	4	3.698	31	176, +28	173, +16	5	3.770	26
S 5	178, +25	174, +16	4	3.990	5	172, +10	171, +03	4	3.972	9	178, +08	177, +01	4	3.909	16
S 6	180, +04	179, −03	6	5.769	15	174, +07	173, −03	3	2.976	14	166, +45	157, +34	3	2.685	54
All S samples	175, +29	172, +17	38	34.433	8	178, +22	175, +11	25	21.533	12	182, +28	178, +18	25	20.452	14
All S sites	174, +28	171, +16	6	5.856	12	178, +20	175, +11	6	5.835	12	182, +31	177, +21	6	5.657	18
Blacks Harbour															
BH1	172, +04	192, +49	9	8.905	6	171, +02	189, +48	5	4.835	16	178, −03	193, +40	4	2.898	70
BH2	156, +25	180, +74	8	6.254	31	150, +01	151, +51	7	6.091	25	152, −07	154, +44	3	2.933	22
All BH samples	166, +13	189, +59	17	14.784	14	160, +01	169, +51	12	10.682	16	165, −04	174, +44	7	5.638	32
Both BH sites	164, +15	189, +62	2	1.949	—	161, +01	171, +51	2	1.955	—	165, −05	174, +44	2	1.934	—
Conglomerates															
C 1	142, +75	153, +63	6	5.730	16	168, +16	168, +03	6	4.926	35					
C 3	156, +27	153, +11	3	2.802	41	154, −12	158, −28	3	2.295	—					
All C1 + C3 samples	151, +60	153, +46	9	7.790	22	163, +07	165, −07	9	6.981	30					
Both C1 + C3 sites	153, +51	153, +37	2	1.802	—	161, +02	163, −12	2	1.923	—					
C 2	161, +46	152, +27	4	1.381	—	185, +79	145, +63	4	0.994	—					
Lavas															
L 1	189, +57	179, +51	8	7.417	17	185, −05	186, −11	4	3.751	28					
L 2	016, +76	078, +69	7	3.327	73	163, +48	154, +33	5	4.456	30					
All L1 + L2 samples	188, +71	164, +63	11	10.006	28	175, +24	170, +13	9	7.341	26					
Both L1 + L2 sites	184, +81	147, +69	2	1.838	—	176, +22	171, +11	2	1.787	—					
L 3	83, +67	114, +60	5	4.747	20	Random at P = 0.05 at 400 °C.									
Baked sediment															
K	166, +25	162, +10	3	2.984	11	166, +36	160 +21	3	2.945	21					

D = the declination clockwise from true north; I = the inclination positive downwards; N = the number of samples; R = the length of the resultant vector, giving each sample unit weight; α = the half-angle of the cone of confidence at a probability P = 0.05. (Fisher 1953). Subscripts h and b refer directions respectively to the horizontal plane and the bedding plane at the site.

TABLE III

Tilt test, St. Andrews – Blacks Harbour

	w.r.t. Horizontal	w.r.t. Bedding
NRM	16°	48°
500 °C	25°	40°
600 °C	39°	23°

The figures are the differences in degrees, between the mean direction at St. Andrews and that at Blacks Harbour, giving each red beds site unit weight.

and after folding respectively. There are no systematic direction changes at St. Andrews and a separation of *A* and *B* components cannot be made. At Blacks Harbour, however, direction changes do occur upon heating, the *B* component being the more stable. However, after the maximum possible thermal demagnetization at 600 °C (at 650 °C directions became random), the *B* direction is not yet reached. Thermal cleaning is incomplete because the stability spectra of *A* and *B* overlap. Clearly *accurate* estimates of the directions of *A* and *B* are not possible but certain limits are set by the observations. The NRM direction at St. Andrews sets an upper limit for the inclination of the *A* component of +28° and that at Blacks Harbour a lower limit of +13° (Fig. 4) both with respect to horizontal. The directions after thermal cleaning (600 °C) at the two localities set limits to the inclination of the *B* component of about +18° and +44° respectively with respect to bedding. The declinations of both components are similar (Table II). Thus the *A* and *B* components have inclinations and declinations that do not differ, within the accuracy of the present analysis, their best estimate being 175°, +23°. The error associated with this estimate cannot be formally calculated, but subjectively we suggest an error of about 15°. Within this accuracy the field has not changed from the time of acquisition of the *B* component before the folding (which is considered to be earlier than Westphalian D in age), and the time of acquisition of the *A* component after folding.

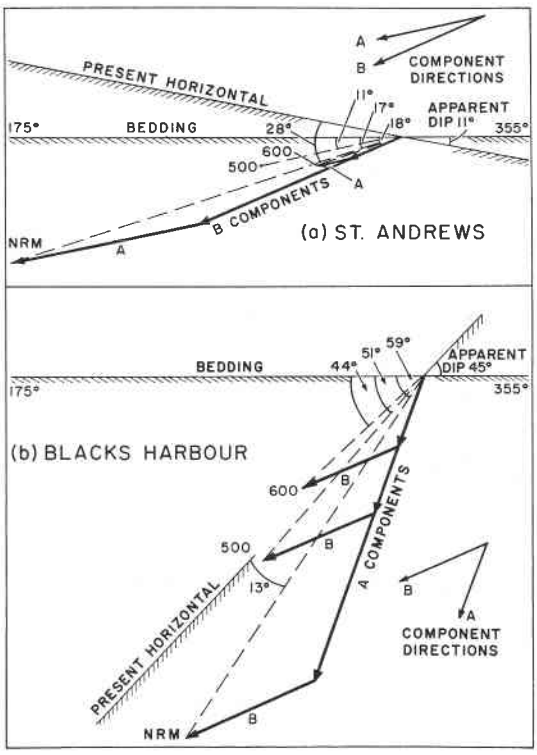


FIG. 4. Diagrammatic representation and interpretation of the effect of thermal cleaning on the mean vectors from (*a*) St. Andrews and (*b*) Blacks Harbour.

Polished sections from Blacks Harbour contain grains of both magnetite and hematite, in addition to the red staining which is presumably Fe_2O_3, whereas in the polished sections from the St. Andrews red sediments magnetite grains are either absent or very subordinate to hematite. At first sight a simple correlation between the *A* component and magnetite, and the *B* component and hematite appears to be possible, the large *A* component at Blacks Harbour resulting from the magnetite there. Such a simple picture cannot be substantiated, however, because from these data, it cannot be established that all of the *A* component at Blacks Harbour has been removed after demagnetization at 600 °C, which is above the Curie temperature of magnetite.

Interpretation

It has been shown that the igneous rocks from the Perry Formation are generally unstable and that the sediments have a rather complex magnetic history. The *B* component of magnetization was acquired before folding. It may be contemporaneous with deposition in the Late Devonian Epoch, or it may be of Mississippian or Early Pennsylvanian age (pre-Westphalian D). It is poorly defined, but a rough estimate given above indicates that the pole was then at about 32° N, 118° E. This is close to the poles for the Upper Silurian Bloomsburg Formation (32° N, 102° E, Roy *et al.* 1967) and the Mississippian Maringouin Formation (34° N, 117° E,

Roy and Robertson 1968). These results are consistent with the idea that the mean field in eastern Canada did not change greatly between Late Silurian and Pennsylvanian times. The *A* component, acquired after folding, is more steeply inclined than the late Pennsylvanian and Permian field in eastern Canada (Roy 1963; 1966); this would suggest that it was acquired before then, probably in early Pennsylvanian time. A burial magnetization of viscous origin (Chamalaun and Creer 1964; Irving and Opdyke 1965) seems possible.

Results from the Perry Formation were previously obtained by Black (1964); our mean direction is not significantly different from his sediment results (174, +20, $\alpha_{95} = 9$). Our thermal demagnetization indicates that the low precision may be due to the presence of two components. In Black's work, the specimens were demagnetized in a maximum peak alternating field of 300 Oe, which was much too low to detect differences in the high stability components. His low precision ($k < 10$) may also be attributed to the presence of two components.

Black used his paleomagnetic data from the Perry Formation to support the hypothesis of a 30° rotation of Newfoundland. Such a small rotation can only be clearly shown by results of high accuracy. Rotation would be shown by differences in declination (D) between beds of the same age in Newfoundland and New Brunswick. His Cambrian data (Bradore Formation of Newfoundland and Ratcliffe Brook Formation of New Brunswick) are not significantly different (the ovals of confidence at $P = 0.05$ intersect). Apparent differences in Devonian (Clam Bank Formation of Newfoundland and Perry Formation of New Brunswick) declination may or may not be significant depending on the error associated with the Perry Formation data (our estimate is 15°, Black's is 9°). Moreover, there is an uncertainty in the age of the magnetization of the Perry Formation as explained above (it may be Mississippian). It is therefore doubtful whether rotation of Newfoundland can be firmly established from the present magnetic data. We do not mean to say that rotation did not occur, but that there is in our opinion no clear-cut paleomagnetic evidence for such a contention at present.

Acknowledgments

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